

Lebanese American University



School of
Engineering

COE 322 – Logic Design Lab **Lab Project**

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1. Introduction

The traditional method of taking classroom attendance - calling names, passing around sign-in sheets, or manually counting heads - is time-consuming, error-prone, and poorly suited to modern educational environments. As classrooms grow larger and institutions demand more accurate records, the need for an automated, real-time occupancy tracking system becomes increasingly clear.

This project presents the design and implementation of a Smart Classroom Attendance System built entirely from digital logic components. The system monitors a classroom entrance and exit using directional input signals and maintains an accurate, real-time count of the number of students currently inside. It enforces a maximum occupancy limit of 29 students, operates under a 10-minute session timer, and provides the instructor with the ability to manually lock the door at any time.

The design is grounded in the core principles of digital logic design studied throughout COE 322. A Mealy-type Finite State Machine governs the entry and exit control logic, a 5-bit synchronous up/down counter tracks occupancy, and a BCD countdown timer manages the session window. All three modules are implemented from first principles using JK and D flip-flops, standard logic gates, and 74-series integrated circuits.

The system provides real-time visual feedback through LED indicators. A green LED signals open access, a red LED signals restricted access, and an orange LED warns when half the session time or half the classroom capacity has been reached. Additional LED lights are used to visually represent important counter and timer output states during testing and operation.

The entire design was first developed analytically using state tables, K-maps, and Boolean minimization, then implemented and simulated in Altera Quartus II, and finally assembled on a virtual breadboard for functional verification.

2. Components and Equipment Used

2.1 Logic and Timing ICs

The following table combines matching components from the FSM module, the main counter module, the 10-to-0 counter module, and the additional support components used in the final breadboard implementation.

Table 1-Logic and Timing ICs

Component	Part Number	Function	Quantity
Hex inverter	74LS04	NOT gates for FSM, timer, reset, and support logic	4
Quad 2-input AND	74LS08	FSM logic, counter gating, timer logic, and LED/support logic	12
Triple 3-input AND	74LS11	Multi-input AND terms used mainly in FSM and counter logic	2
Quad 2-input OR	74LS32	Combining FSM, counter, timer, and support logic terms	3

Component	Part Number	Function	Quantity
Dual D flip-flop	74LS74	FSM state storage using Q1 and Q0	1

Dual JK flip-flop	74LS76	Counter and 10-to-0 timer flip-flops	5
Timer IC	555 Timer	Clock pulse generation for the timer circuit	1

2.2 Passive Components

Table 2- Passive Components

Component	Specification	Function	Quantity
Resistor	300 ohm	LED current limiting	10
Resistor	1 kohm	Support resistance for signal conditioning or output limiting	1
Resistor	10 kohm	555 timer or switch support network	1
Resistor	75 kohm	555 timer frequency-setting network	1
Resistor	100 kohm	555 timer frequency-setting network	1
Capacitor	10 uF	555 timer and support capacitance	3

2.3 Output Devices

Table 3-Output Devices

Component	Specification	Quantity
LED lights	Standard LEDs for counter/timer output indication	10
Colored LEDs	Red, green, and orange indicators	3
Buzzer	5V buzzer	1

2.4 Power, Control, and Assembly

Table 4-Power, Control, and Assembly

Component	Specification	Quantity
Voltage source	5V DC	1
Switches	Manual control inputs	4
Breadboards	Standard breadboards	8
Soldering breadboard	Permanent/prototype soldering board	1
Male-to-male wires	Jumper wires	About 300

2.5 Software

Table 5-Software Used

Tool	Purpose
Altera Quartus II	HDL/schematic entry, synthesis, simulation
Digital Circuit Virtual Builder	Virtual breadboard implementation and testing

3. Analysis

3.1 System Architecture

FSM Module - A Mealy-type finite state machine with 4 states (IDLE, ENTER-HOLD, EXIT-HOLD, LOCKED) implemented using two D flip-flops. The FSM takes six inputs (R, L, E, X, MAX, ZERO) and produces three outputs (UP, DOWN, TIMER_RST) that control the counter and timer modules.

5-bit Up/Down Counter Module - A synchronous counter built from five JK flip-flops that maintains the current student occupancy count from 0 to 29. It receives UP_gated and DOWN_gated pulses from the FSM and generates MAX and ZERO feedback signals.

BCD Countdown Timer Module - A synchronous countdown timer built from four JK flip-flops counting from 10 down to 0, driven by a 1 Hz clock from the 555 timer circuit. It generates TIMEOUT, used to automatically lock the door, and HALF_TIME, used to trigger the orange LED.

Output Logic Module - Combinational logic deriving the three main LED indicator signals and driving the LED output lights from the FSM state, counter value, and timer value.

3.2 FSM Design

The FSM is a Mealy machine with 4 states encoded in 2 bits (Q1, Q0):

Table 6 - FSM State Encoding

State	Q1	Q0	Meaning
S0 - IDLE	0	0	Door open, waiting
S1 - ENTER-HOLD	0	1	Entry in progress
S2 - EXIT-HOLD	1	0	Exit in progress
S3 - LOCKED	1	1	Door locked

The state table was constructed using all 6 inputs (R, L, E, X, MAX, ZERO) with R having highest priority. After K-map minimization over 5 variables (Q1, Q0, L, E, X) with MAX and ZERO applied as guard terms, the minimized Boolean equations are:

$$\begin{aligned}D1 &= [L + E'.X.(Q1.Q0)'] \cdot R' \cdot ZERO' \\D0 &= [L + Q1'.E.X'] \cdot R' \cdot MAX' \\UP &= Q1'.E.X' \cdot R' \cdot MAX' \\DOWN &= Q0'.E'.X \cdot R' \cdot ZERO' \\TIMER_RST &= Q1.Q0.L' \cdot R'\end{aligned}$$

The ENTER-HOLD and EXIT-HOLD states (S1 and S2) exist specifically to prevent one held button press from producing multiple count pulses. The FSM only counts once per complete press-and-release cycle.

3.3 Counter Design

The 5-bit up/down counter uses JK flip-flops because their built-in toggle mode ($J=K=1$) directly maps onto the counter toggle pattern, eliminating the XOR gates that would be required with D flip-flops. This saves approximately 2 chips compared to the D flip-flop alternative.

The JK excitation equations for each bit follow the carry/borrow toggle pattern:

$$J0=K0 = UP_gated + DOWN_gated$$

$$\begin{aligned}
J1=K1 &= (\text{UP_gated.Q0}) + (\text{DOWN_gated.Q0'}) \\
J2=K2 &= (\text{UP_gated.Q1.Q0}) + (\text{DOWN_gated.Q1'.Q0'}) \\
J3=K3 &= (\text{UP_gated.Q2.Q1.Q0}) + (\text{DOWN_gated.Q2'.Q1'.Q0'}) \\
J4=K4 &= (\text{UP_gated.Q3.Q2.Q1.Q0}) + (\text{DOWN_gated.Q3'.Q2'.Q1'.Q0'})
\end{aligned}$$

Overflow and underflow protection is provided by gating the UP and DOWN signals before they reach the counter:

$$\begin{aligned}
\text{UP_gated} &= \text{UP} \cdot \text{MAX}' \cdot \text{R}' \\
\text{DOWN_gated} &= \text{DOWN} \cdot \text{ZERO}' \cdot \text{R}'
\end{aligned}$$

MAX (count = 29 = binary 11101) and ZERO (count = 0 = binary 00000) are derived combinationaly from the counter Q outputs:

$$\begin{aligned}
\text{MAX} &= \text{Q4.Q3.Q2.Q1'.Q0} \\
\text{ZERO} &= \text{Q4'.Q3'.Q2'.Q1'.Q0'}
\end{aligned}$$

3.4 Timer Design

The BCD countdown timer uses four JK flip-flops implementing a synchronous BCD down counter. The derived JK equations from K-map analysis are:

$$\begin{aligned}
J0 &= K0 = 1 & (Q0 \text{ always toggles - tie to VCC}) & J1 = Q1'.Q0' \\
J2 &= Q3.Q1'.Q0' & K1 &= Q0' \\
K3 &= Q2'.Q1'.Q0' & K2 &= Q1'.Q0' \quad J3 =
\end{aligned}$$

The timer is loaded to 10 (1010 in binary) on reset via the PR' and CLR' pins of the flip-flops. When TIMER_RST from the FSM goes HIGH (on S3 to S0 unlock transition), the timer reloads to 10 without affecting the student count.

TIMEOUT and HALF_TIME are derived as:

$$\begin{aligned}
\text{TIMEOUT} &= \text{Q3'.Q2'.Q1'.Q0'} & (\text{timer} = 0) \\
\text{HALF_TIME} &= \text{Q3'.(Q2' + Q1'.Q0')} & (\text{timer} \leq 5)
\end{aligned}$$

TIMEOUT is OR'd with the physical lock switch before reaching the FSM L input, ensuring the door locks automatically when time expires.

3.5 Output Logic

LED signals:

$$\begin{aligned}
\text{GREEN} &= \text{FSM_Q1'.FSM_Q0'.MAX'.TIMEOUT'.L'} & (\text{IDLE, not full, not timed out, not locked}) & \text{RED} = \\
&(\text{FSM_Q1.FSM_Q0}) + \text{MAX} + \text{TIMEOUT} & & (\text{locked, full, or timed out}) \\
\text{ORANGE} &= \text{HALF_CAP} + \text{HALF_TIME} & & (\text{half capacity or half time reached}) \\
\text{HALF_CAP} &= \text{Q4} + (\text{Q3.Q2.Q1}) & & (\text{count} \geq 14)
\end{aligned}$$

LED output implementation:

The green, red, and orange LEDs provide the main system status feedback. The green LED turns ON when the classroom is available for entry, the red LED turns ON when access is restricted, and the orange LED turns ON when either half the classroom capacity or half the session time has been reached.

The remaining LED lights are used to make the counter and timer states visible during testing. Each LED is connected through a current-limiting resistor so that the binary output states can be observed directly from the breadboard circuit.

4. Paper Design

The paper design phase was the first stage of the implementation process. In this stage, all modules of the Smart Classroom Attendance System were designed manually using digital logic analysis before being implemented in Quartus and on the breadboard.

State tables, excitation tables, Karnaugh maps, and Boolean minimization techniques were used to derive the required logic equations for each module. The final minimized equations were then converted into gate-level paper schematics using standard 74-series logic components.

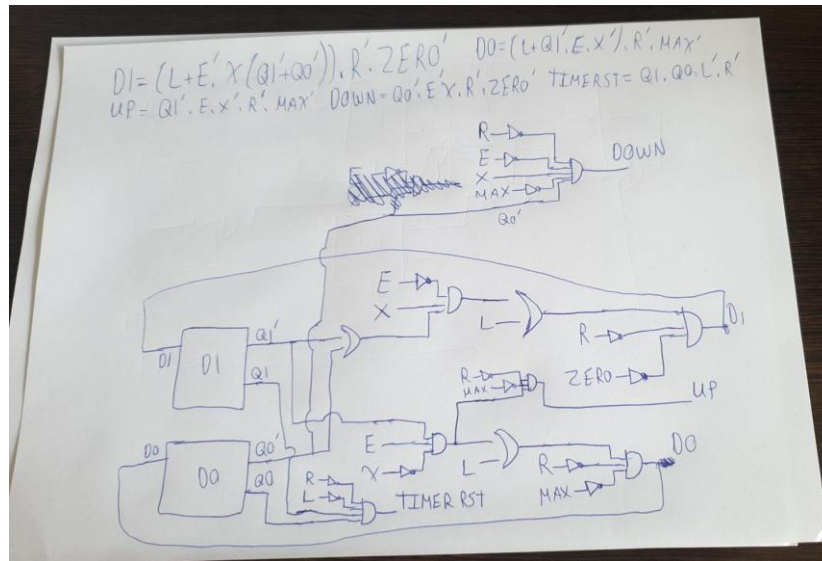


Figure 1- FSM Paper Design

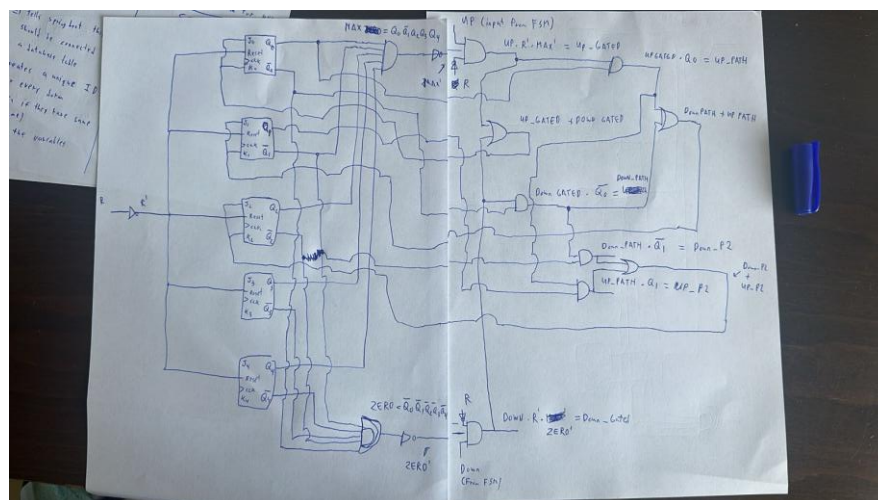


Figure 2 - Counter Paper Design

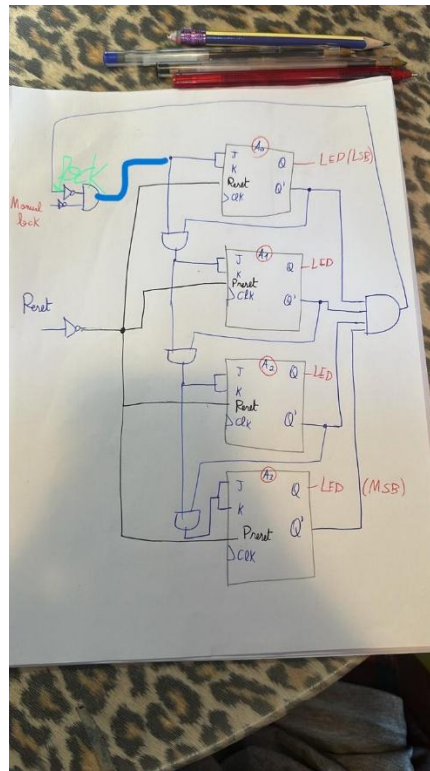


Figure 3- Countdown Timer

5.Quartus Design and Analysis

After completing the paper design stage, the system modules were implemented in Altera Quartus II for verification and simulation. Each module was first built separately as an independent schematic block, including the FSM, the 5-bit up/down counter, the countdown timer, and the output logic module. The modules were then connected together to form the complete Smart Classroom Attendance System.

Functional simulations were performed to verify correct operation under different input conditions. The FSM simulation confirmed proper transitions between the IDLE, ENTER-HOLD, EXIT-HOLD, and LOCKED states. The counter simulation verified correct increment and decrement behavior while preventing overflow above 29 and underflow below 0. The timer simulation confirmed proper countdown operation from 10 to 0 and successful generation of the TIMEOUT signal.

The Quartus simulations also verified that:

- Only one count pulse is generated per entry or exit operation.
- The system locks automatically when the timer expires or when maximum occupancy is reached.
- The timer resets correctly when unlocking the door without affecting the student count.
- The LED outputs respond correctly to all system states.

The delay calculations are shown under each simulation

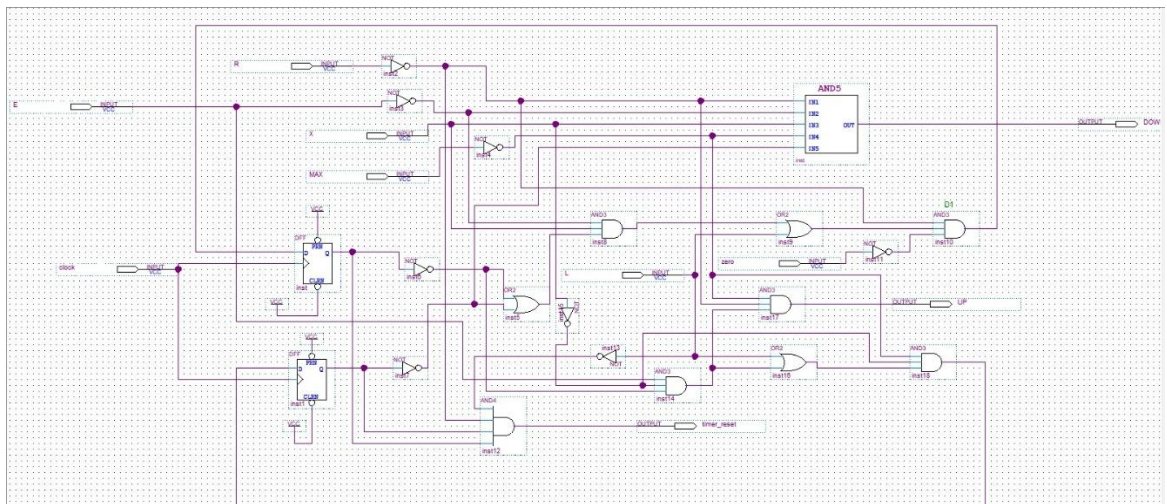


Figure 4 FSM Quartus

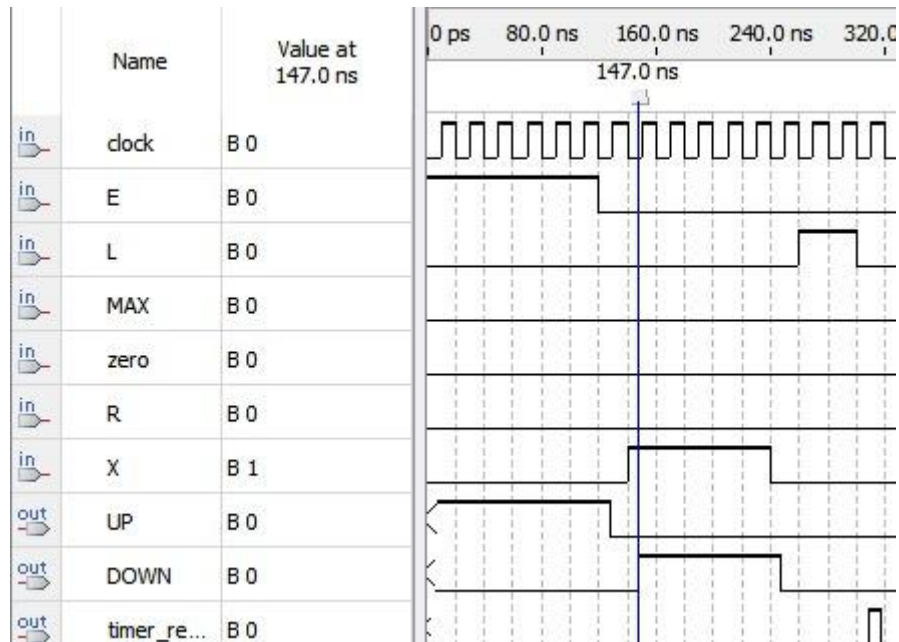


Figure 5-Simulation of FSM on quartus

The FSM was shown to have a 7ns delay

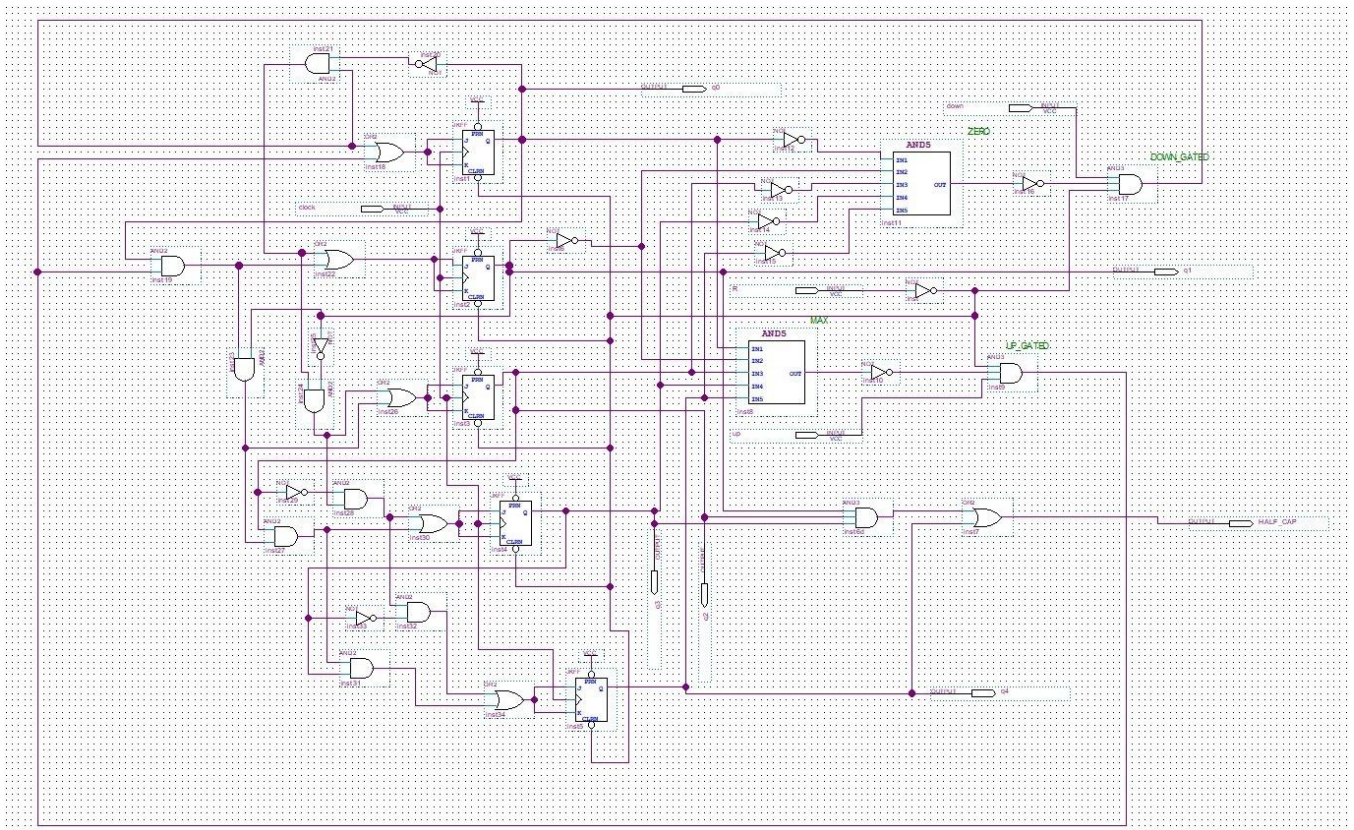
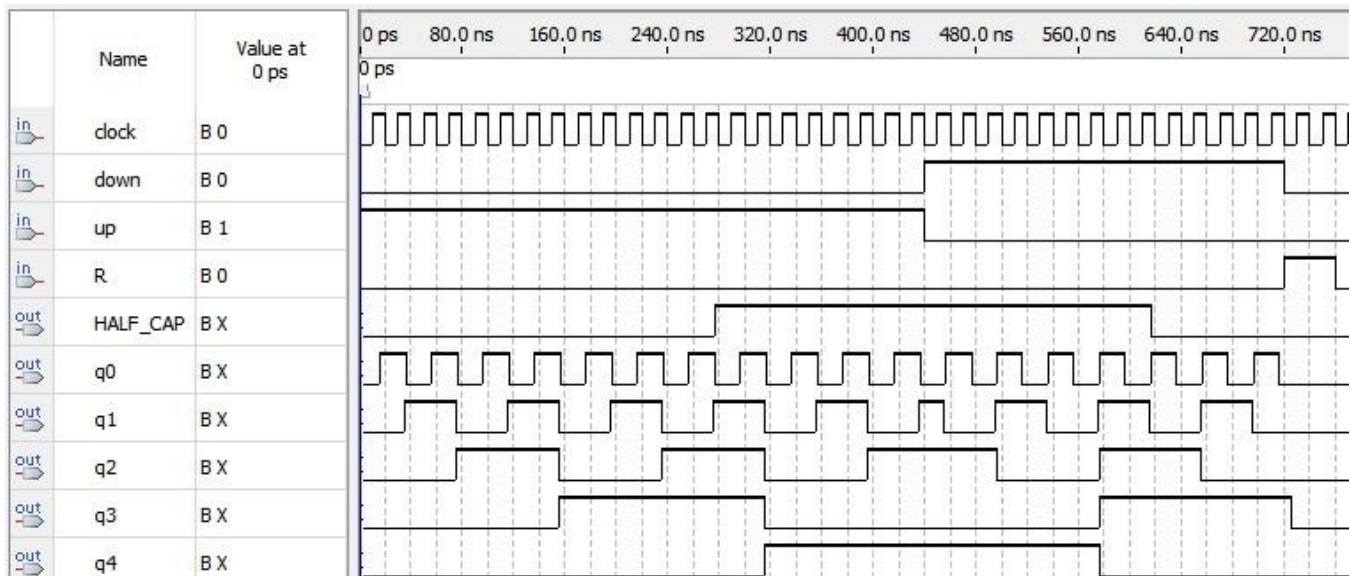


Figure 6 - Counter Quartus



-Figure 7--Simulation of counter on quartus

The counter was shown to have a 2ns delay

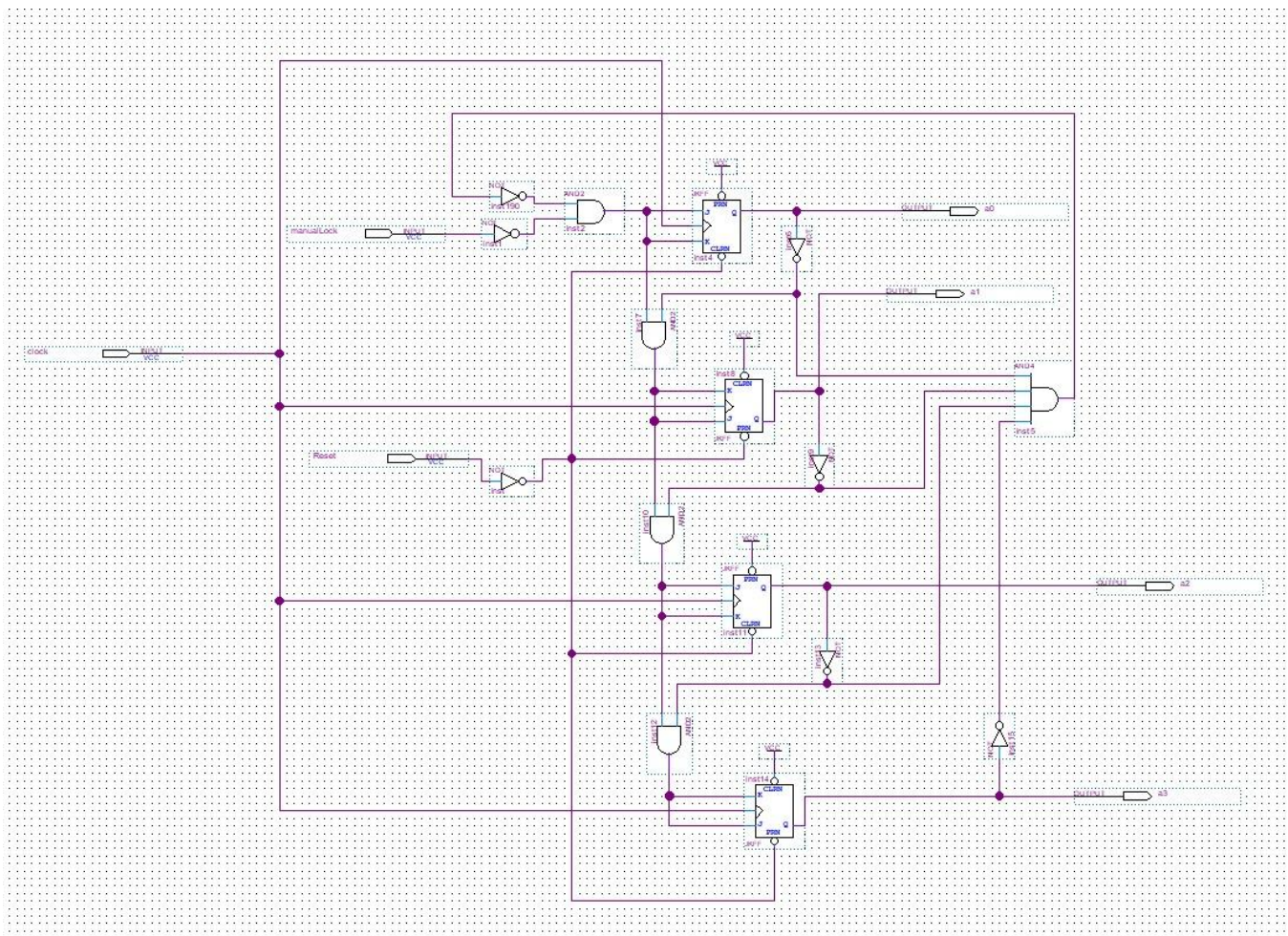


Figure 8-Timer on quartus

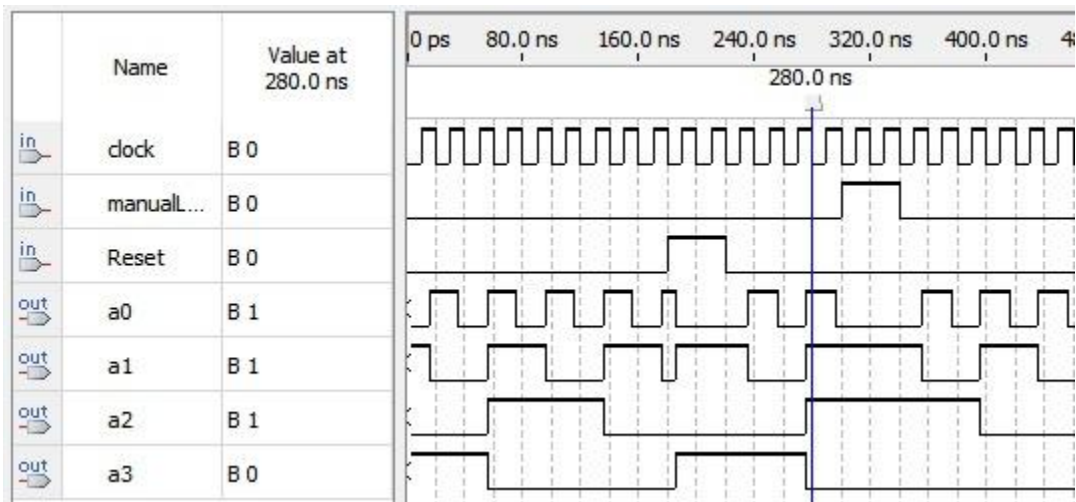


Figure 9--Simulation of timer on quartus

The timer was shown to have a 6ns delay

6.Breadboard Design and Analysis

After successful Quartus simulation, the design was implemented on a virtual breadboard using Digital Circuit Virtual Builder software. The breadboard implementation reproduced the real hardware connections using 74-series ICs, LEDs, switches, resistors, capacitors, and the 555 timer circuit.

The design was implemented in separate modules to simplify testing and debugging. The FSM, counter, and timer circuits were tested independently before integrating the complete system. LED indicators were connected to the counter and timer outputs to visually monitor system operation during testing.

The virtual breadboard implementation confirmed that:

- The occupancy counter increments and decrements correctly.
- The FSM correctly handles entry, exit, and lock conditions.
- The timer counts down properly and activates automatic locking at timeout.
- The green, red, and orange LEDs operate according to the required specifications.
- The reset functions correctly restore the system to its initial state.

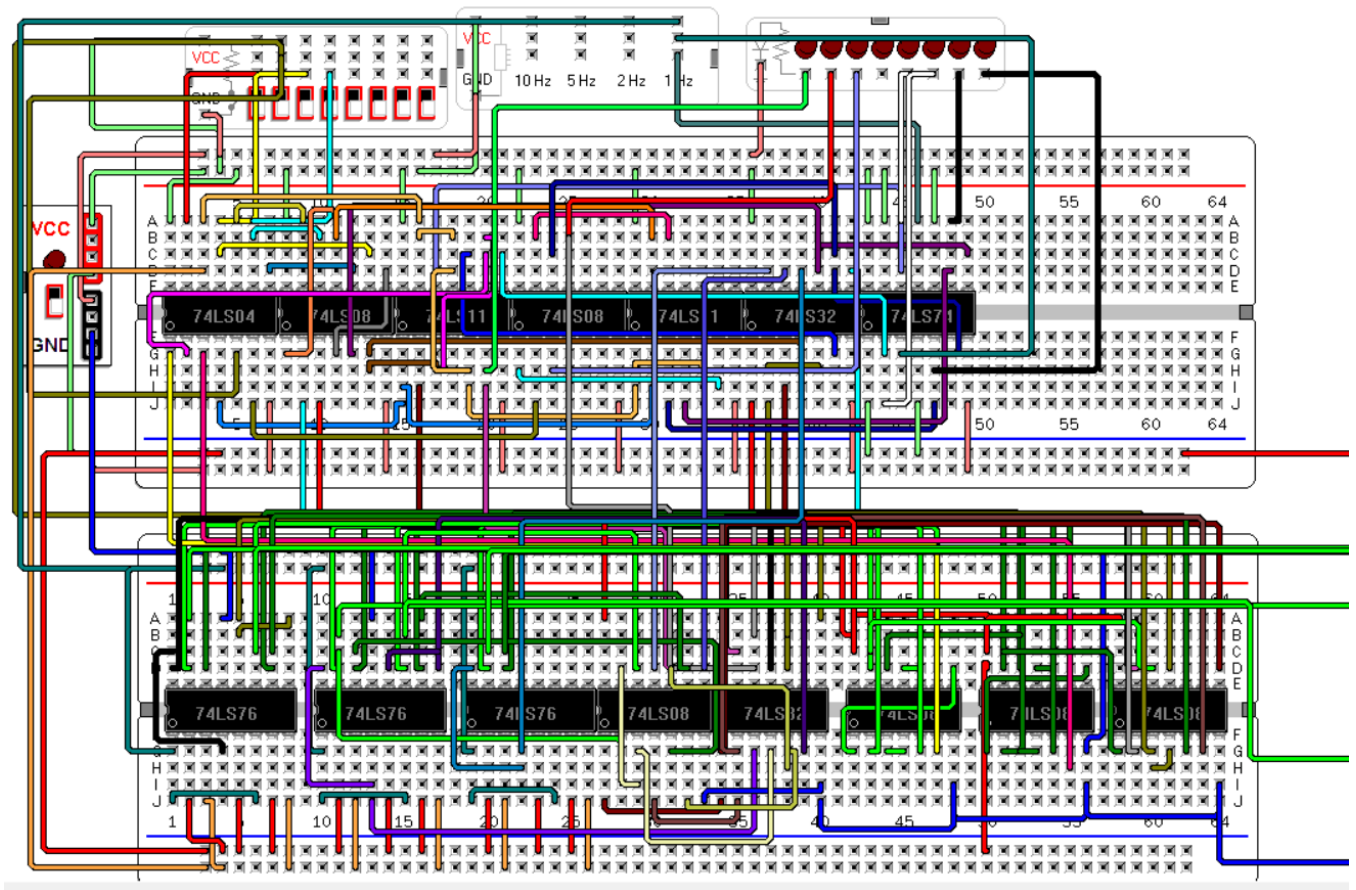


Figure 10-Virtual breadboard of counter

Next the counter working states will be shown below:

First incrementing past 29 does not work

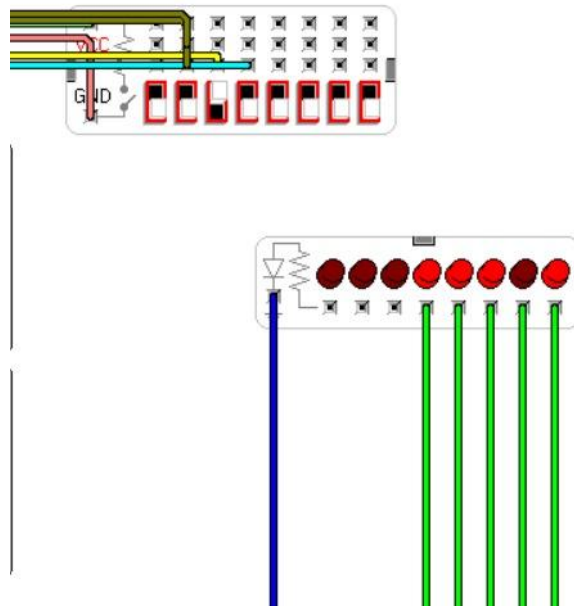


Figure 11-Counter state past 29 incrementing

Also decrementing past zero does not work:

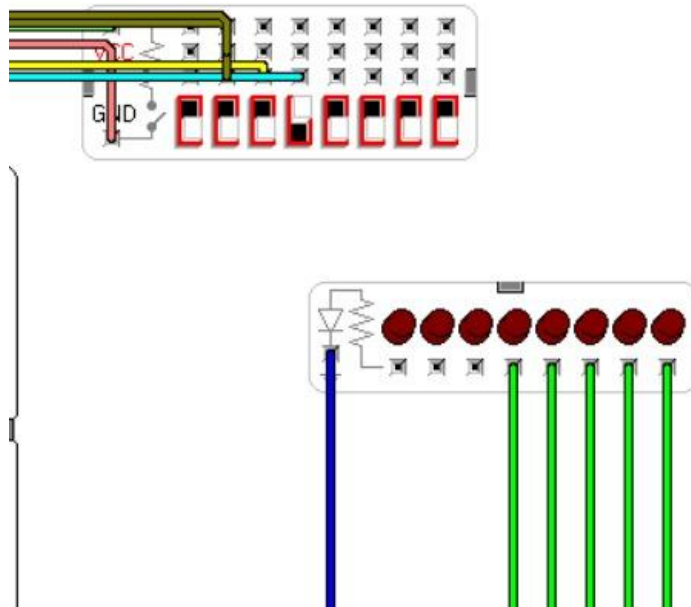


Figure 12-Counter state past 0 decrementing

Next, when the counter is in the reset state, it goes back to zero and no other inputs work:

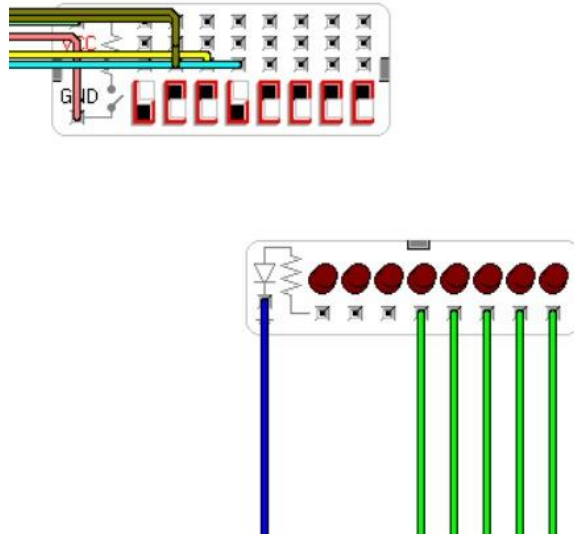


Figure 13-Counter in reset state

Finally when the counter is in a locked state, decrementing does not work

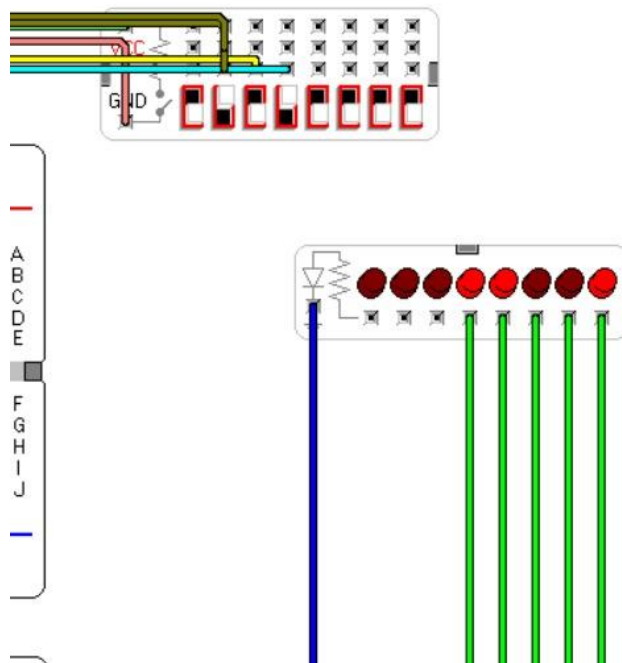


Figure 14-Counter in locked state

Next the design of the FSM will be shown

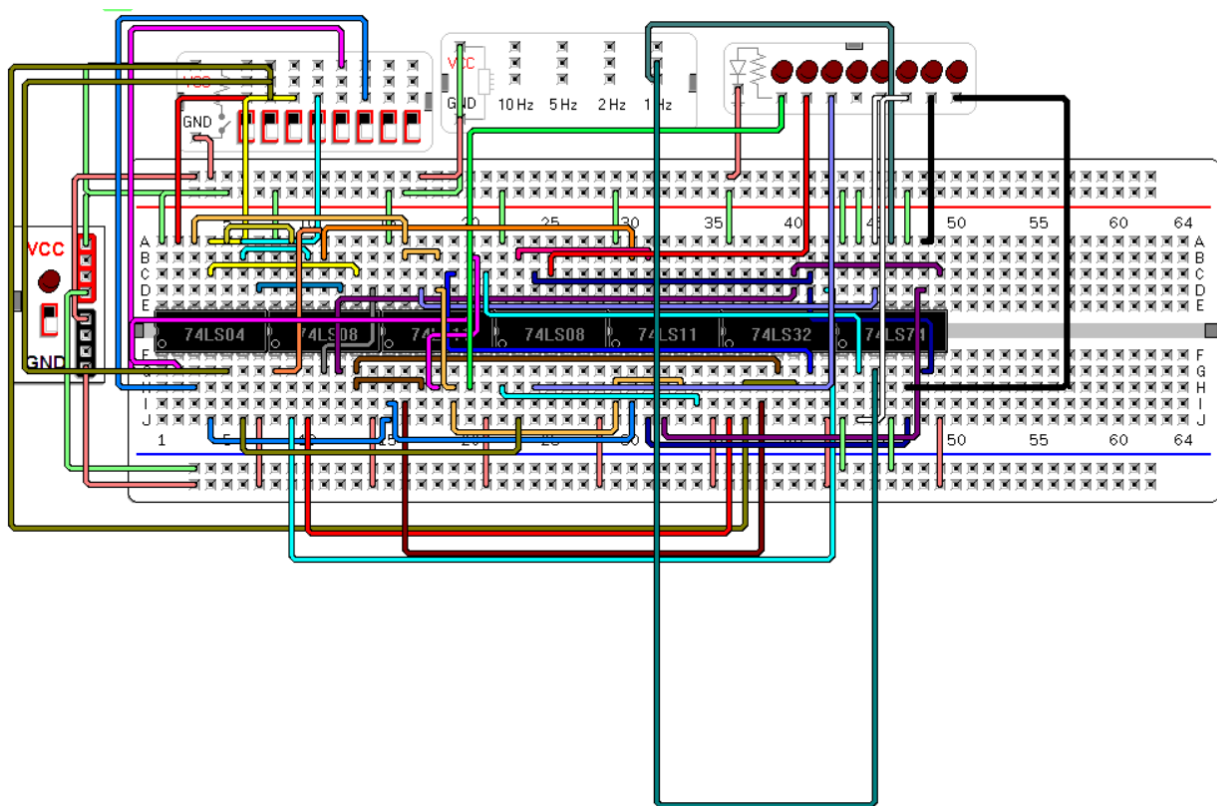


Figure 15-FSM on virtual breadboard

Next, the design of the counter will be shown

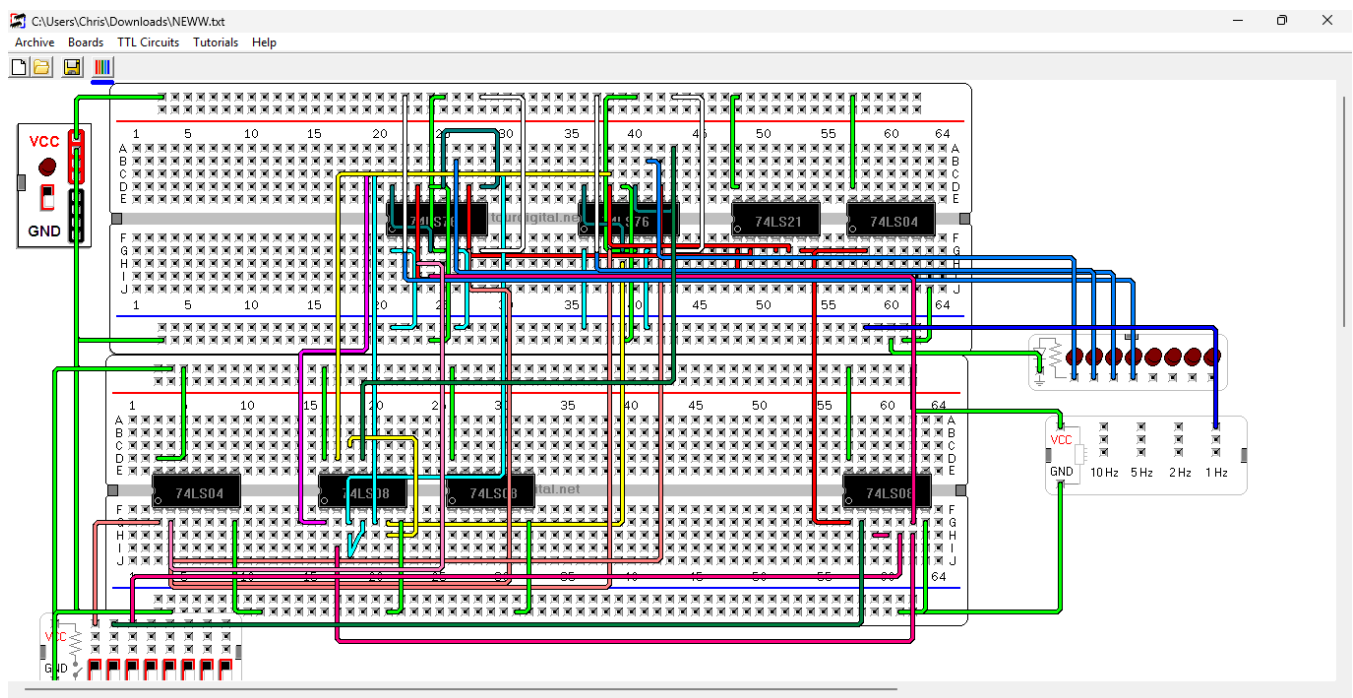


Figure 16-Timer on virtual breadboard

Next screenshots of the timer working will be provided:

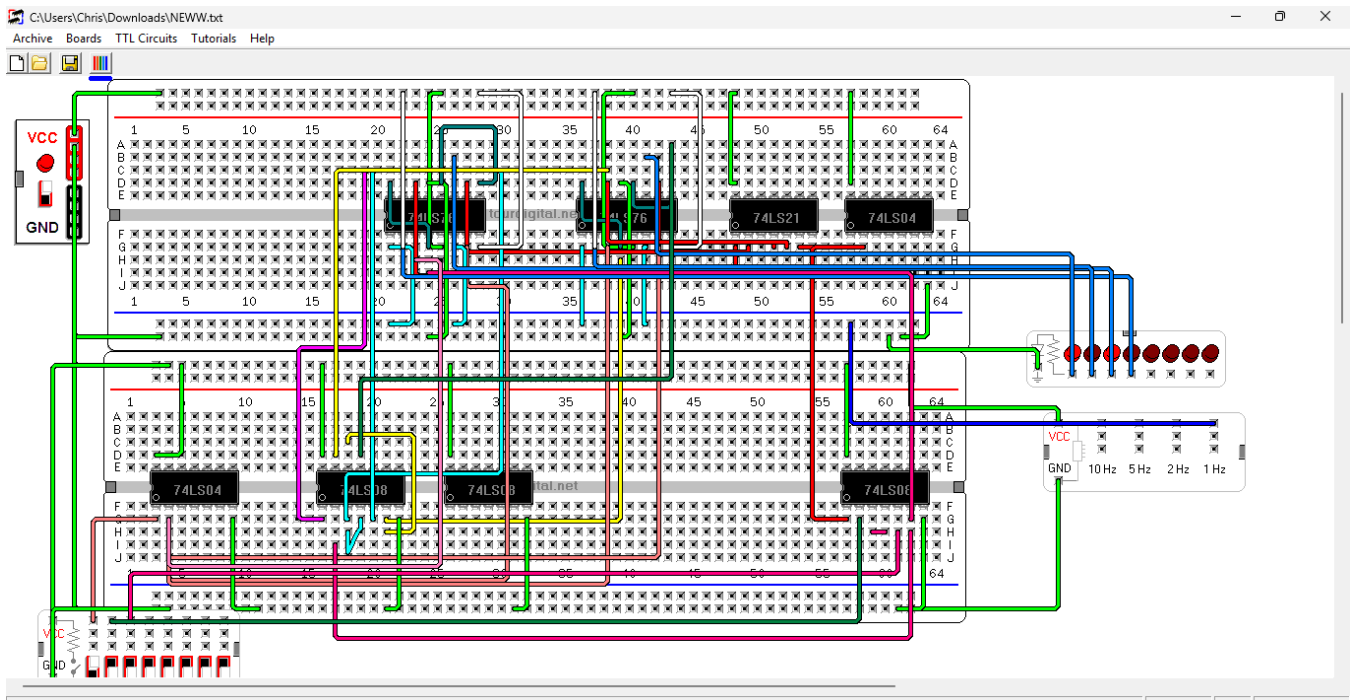


Figure 17-Timer when reset is high

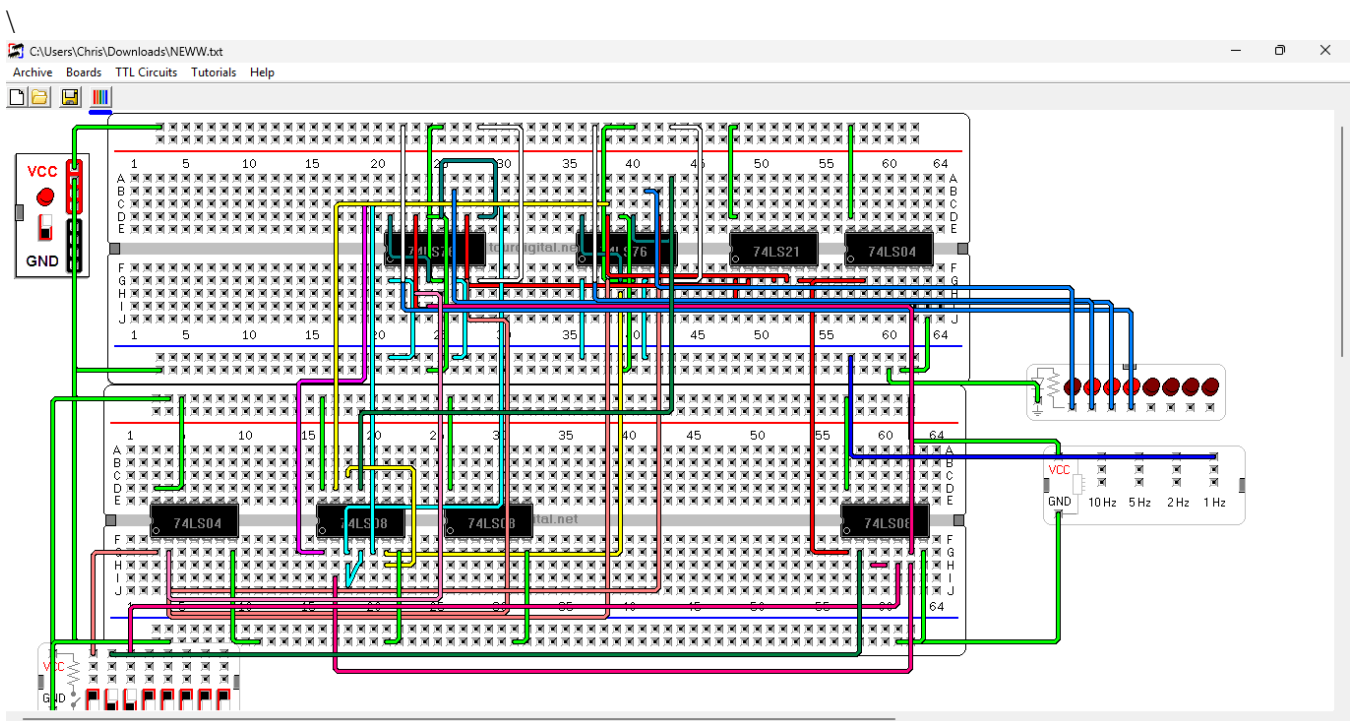


Figure 18-Timer counting from 10 to 0 when it is at 7

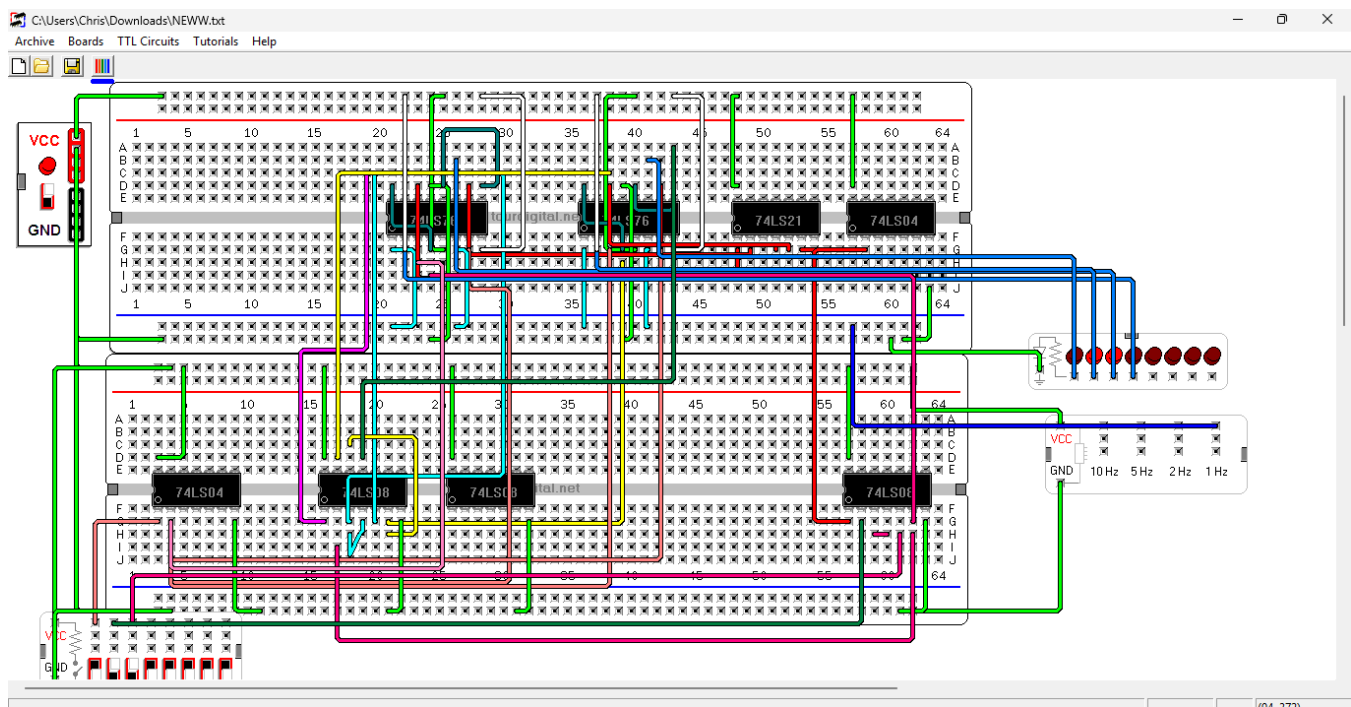


Figure 19--Timer counting from 10 to 0 when it is at 6

7.Financial Study:

In this part a financial analysis and report will be done for the project's components

Below is attached a receipt of all the components bought for the project

INCOTEL ELECTRONICS DORA, ST. JOSEPH STR., FEGHALI CENTER, BEIRUT, LEBANON TEL. 961 1 240003 - FAX. 961 1 252276					
VAT # 70973-601			Sales offer # 01505		
M/s CASH SALES			Date 09/05/2026		
			Page 01/01		
Item #	Description	Quantity	Unit Price	(-%)	Total US
74HCT04	6 INVERTER	3	1,300		3.90
74HCT08	4X2 AND HIGHT SPEED CMOS -	11	1,200		13.20
74HCT11	3X3 AND.	2	1,000		2.00
74HCT32	4X2 OR	2	1,250		2.50
74HCT74	DUAL D-FU/FLOP H-SPEEDCMOS (HCTLS)	1	1,000		1.00
74LS76	2 JK FL/FP RESCL	5	3,000		15.00
LD59	LED GREEN 5MM STANDARD	1	0,100		.10
LD65	LED ORANGE 5MM STANDARD	1	0,100		.10
LD91	LED RED 5MM STANDARD	11	0,100		1.10
10MF/25VA	LELON-5.0x11-REA100M1E0511-85G-ROHS	3	0,150		.45
FJ65	FLEXIBLE BREADBOARD JUMPER 2.54mm 65PCS	3	3,000		9.00
DHJ3	DUPONT HEAD JUMPER MALE/FEM-2.54mm	1	3,000		3.00
PF801	PERFORATED PCB SS 9x15cm 1440 Points	1	1,500		1.50
MB-102	BREADBOARD ROUND HOLE 840 Tit POINTS-CHINA	8	2,750		22.00
DC026	DC JACK M TO TERMINAL 2.1mm	1	1,000		1.00
SW4009	XXXX TIRB24	4	0,500		2.00
BZZER 1	BZZER 5V PASSIVE PIEZO SOUND 5VDC	1	1,000		1.00
Total Qty =		59	Gross		78.85
			Discount		0.00
			Net		78.85
VAT LL		776,278	VAT 11%		8.67
			Net total		87.52
ONLY EIGHTY SEVEN US DOLLAR & 52% .					
			Net LL		7,833,353
Remark.					

Figure 20-Receipt of all equipment

The total cost of the project was optimized thanks to multiple cost optimization techniques shown below:

Cost Optimization Techniques

Several design choices were made specifically to reduce hardware cost and chip usage:

- JK flip-flops were selected for the counter and timer modules because they directly support toggle operation, eliminating the need for additional XOR gates.
- The complementary outputs (Q') available directly from the 74LS76 flip-flops were reused instead of adding external inverter ICs.
- Shared logic expressions were reused across multiple outputs to reduce the total number of AND and OR gates required.
- Standard TTL ICs were used because they are inexpensive and widely available.
- The system was divided into reusable modules (FSM, counter, timer, output logic), simplifying debugging and reducing redesign cost.

8. Problems Faced

8.1 CLR' Pin Misconnection on Flip-Flops

During breadboard implementation, both the counter and FSM flip-flops were found to be permanently held in the reset state. The cause was identified as the CLR' pin being inadvertently connected to the reset switch directly rather than through an inverter. Since CLR' is active LOW, having it connected to R (active HIGH) meant that whenever the reset switch was LOW (which is its normal idle state), CLR' was also LOW and the flip-flop was continuously clearing. The fix was to route all CLR' connections through an inverter to produce R', which goes LOW only when the reset switch is actually pressed.

8.2 D to Q Transfer Not Occurring on FSM Flip-Flops

After fixing the CLR' issue, the FSM D flip-flops were still not transferring the D input value to Q on the clock edge. This was traced to the PR' (preset) pin being unconnected and floating. A floating PR' pin on the 74LS74 was randomly asserting, holding Q HIGH regardless of the clock or D input. The fix was to tie both PR' pins to VCC to disable the preset function.

8.3 TIMER_RST Not Activating

The timer reset signal was not triggering correctly when the door was unlocked from the LOCKED state. This was found to be caused by the system never reaching S3 in the first place - the reason being the CLR' issue described above which prevented the flip-flops from ever reaching $Q1=Q0=1$. Once the CLR' wiring was corrected, the FSM was able to reach S3, and TIMER_RST activated correctly on the S3 to S0 transition.

8.4 State Encoding Conflict in State Table

During the initial state table construction, the ZERO condition was not included in the blocking logic for S0 to S2 transitions and the MAX condition was not included in the S0 to S1 blocking logic. This allowed the counter to attempt decrementing from 0 and incrementing past 29 in simulation. These were corrected by including MAX and ZERO as additional input columns in the state table and deriving the guard terms that were then multiplied into the D0 and D1 equations respectively.

9 Key Design Advantages

9.2 JK Flip-Flops for the Counter - Eliminating XOR Gates

The most significant efficiency gain in this design came from choosing JK flip-flops for the counter and timer modules instead of D flip-flops. A D flip-flop counter requires one XOR gate per bit to implement the toggle condition ($D = Q \text{ XOR } T$), which for a 5-bit counter means several additional gates and IC packages. The JK flip-flop eliminates all of these since setting $J=K=1$ directly implements toggle. This saved approximately 2 chips in the counter alone and simplified the schematic considerably.

9.3 Free Q' Signals from Flip-Flop Output Pins

All 74LS76 JK flip-flops provide both Q and Q' directly as output pins. By taking Q1', Q0', Q2', Q3', and Q4' directly from these pins rather than adding external inverter gates, the design eliminated the need for an entire 74LS04 chip that would otherwise be dedicated to counter bit inversions. This is a hardware cost saving that is easy to overlook but meaningful at the IC level.

9.4 Hold States in the FSM Preventing Multiple Counts

Many simpler attendance counter designs trigger a count pulse directly on the rising edge of the entry or exit sensor. The problem with this approach is that if the sensor or switch is held HIGH for multiple clock cycles, the counter increments multiple times for a single physical entry. This design avoids that entirely by introducing the ENTER-HOLD (S1) and EXIT-HOLD (S2) states. The count pulse is only generated on the first cycle of entry into S1 or S2, and holding the switch simply keeps the FSM in the hold state without generating additional pulses. One button press always equals exactly one student counted.

9.5 Automatic Lock on Timer Expiry

Rather than relying on the instructor to manually lock the door when the session timer reaches zero, the design routes the TIMEOUT signal through an OR gate with the physical lock switch before it reaches the FSM L input. This means the door locks automatically the moment the timer reaches zero, with no manual intervention required. The automatic and manual lock mechanisms are completely interchangeable from the FSM's perspective - both simply assert L_effective HIGH.

9.6 Non-Destructive Unlock - Timer Resets Without Losing Count

The specification required that unlocking the door after a manual lock should reset the timer but preserve the student count. This is handled cleanly by the TIMER_RST output of the FSM, which only fires on the S3 to S0 transition (unlock) and connects only to the timer flip-flop PR'/CLR' pins - never to the counter flip-flops. This separation of reset signals ensures that the two independent resets (full system reset via R, and timer-only reset via TIMER_RST) cannot interfere with each other.

9.7 Minimized Chip Count Through Gate Sharing

Shared sub-expressions ($Q1'.E.X'$ and $Q0'.E'.X$) are computed once and wired to multiple output equations (D0 and UP share the first term; D1 and DOWN share the second). This avoids duplicating AND gates that would otherwise appear in each equation independently. Combined with the reuse of inverters on the input bus and Q' signals from flip-flop pins, the total gate count for the FSM combinational logic is kept to approximately 8 chips rather than the 12 or more that an unoptimized implementation would require.

Conclusion

The Smart Classroom Attendance System was successfully designed and implemented using digital logic components and sequential circuit design techniques. The system achieved all required objectives, including tracking the number of students inside the classroom, controlling entry and exit operations, managing a countdown timer, and automatically locking the classroom when the timer expired or maximum occupancy was reached.

Throughout the project, important digital logic concepts such as finite state machines, synchronous counters, Karnaugh map minimization, and timing analysis were applied in a practical way. The design process began with analytical paper design, followed by Quartus implementation and simulation, and finally virtual breadboard testing. The simulations and practical implementation confirmed correct operation of the FSM, counter, timer, and output logic under different conditions.

In addition, several optimization techniques were used to reduce the total chip count and implementation cost, such as using JK flip-flops for toggle operations and reusing shared logic expressions. Despite encountering multiple implementation challenges, all issues were successfully solved, improving both the reliability of the system and the understanding of real hardware behavior.

Overall, the project provided valuable experience in both theoretical and practical digital system design while demonstrating a complete and efficient smart attendance solution.